

Triage Algorithm Overview

Model architecture and validation approach

Architecture

A 3D convolutional backbone for volumetric head CT and a 2D backbone for chest radiography, each followed by a finding specific classification head. Studies are normalised per scanner vendor before inference.

Training data

218,000 studies in total. 61 percent from Indian sites under data sharing agreements, the remainder from public research datasets. Every Indian study was de identified at source before transfer.

Labelling

Each training study carries a label from the original report plus, for 24,000 studies in the validation set, an independent read by two radiologists with a third breaking ties.

Calibration per site

In the first 30 days at a new hospital the model runs in shadow mode. Thresholds are tuned to that site's scanner and protocol mix before any notification is enabled.

Known limitations

Paediatric studies are excluded. Post operative heads are flagged as low confidence. Portable chest films at very low exposure show a measurable drop in specificity, which is why those are held to a higher notification threshold.